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DEPARTMENT OF ROBOTICS AND MECHATRONICS ENGINEERING

Reinforcement Learning for Mars Rover Wheel Entrapment Recovery in Granular Regolith

A Research Project Report

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Abstract

This report presents a reinforcement learning (RL) framework for autonomous recovery of Mars rovers from wheel entrapment in granular regolith. We leverage the Newton physics engine with Material Point Method (MPM) for high-fidelity granular soil simulation, Isaac Lab for articulated robot simulation, and Proximal Policy Optimization (PPO) with recurrent GRU networks for policy training. The system addresses the critical challenge of wheel entrapment faced by planetary exploration rovers — exemplified by NASA's Spirit rover incident in 2009 — by training escape policies in a physically realistic simulation environment.

Our approach features a 29-dimensional proprioceptive observation space incorporating wheel velocities, slip ratios, IMU data, drive torques, and dual-sensor entrapment detection flags. The policy is trained across 64 parallel environments with curriculum learning that progressively deepens wheel sinkage from 8 cm to 15 cm. Preliminary results over 200,000 timesteps on an RTX 5070 Ti demonstrate an 80% escape rate, mean forward velocity of 0.25 m/s, and full curriculum completion. The trained policy exhibits emergent escape behaviors including rocking maneuvers and coordinated wheel–steering actions.

Future work includes scaling training to 2M+ timesteps, developing a CNN-GRU sinkage detection classifier, and deploying the trained policy on a Raspberry Pi 5 via ONNX Runtime for real-time inference on physical hardware.

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1 Introduction

1.1 Problem Statement

Planetary exploration rovers operating on Mars and other celestial bodies face a critical operational challenge: wheel entrapment in granular regolith. When a rover's wheels sink into loose sand or soil, the vehicle loses mobility and may become permanently immobilized. This problem is particularly acute for several reasons:

- **Wheeled rovers with limited articulation**: The rocker-bogie suspension system, while excellent for traversing rough terrain, distributes weight in ways that can cause individual wheels to sink disproportionately in soft soil.
- **Sandy and cratered terrain**: The Martian surface contains extensive dune fields, impact craters filled with fine regolith, and wind-blown sand deposits that present persistent mobility hazards.
- **Long communication delays**: With Earth–Mars communication latencies of 4–24 minutes one-way, real-time teleoperation for recovery is impractical, necessitating autonomous solutions.
- **Extended mission lifetimes**: As missions extend beyond planned durations, rovers inevitably encounter terrain types not anticipated in original mission planning.

Wheel entrapment can be formally characterized by the conjunction of three conditions:

$$\text{Entrapment} \Leftrightarrow (|v_x| < v_{\text{thresh}}) \land \left(\bar{\sigma}_{\text{slip}} > \sigma_{\text{thresh}}\right) \land \left(d_{\text{sinkage}} > 0\right) \tag{1}$$

where v_x is the body-frame forward velocity, $\bar{\sigma}_{\text{slip}}$ is the mean slip ratio across all wheels, and d_{sinkage} is the wheel sinkage depth into the granular medium.

1.2 Problem Definition

The entrapment recovery problem is formulated as a partially observable Markov decision process (POMDP). At each timestep t, the rover receives a proprioceptive observation $\mathbf{o}_t \in \mathbb{R}^{29}$ and must select an action $\mathbf{a}_t \in \mathbb{R}^{10}$ to maximize the expected cumulative reward:

$$J(\pi_\theta) = \mathbb{E}_{\tau \sim \pi_\theta}\left[\sum_{t=0}^{T} \gamma^t R(\mathbf{s}_t, \mathbf{a}_t)\right] \tag{2}$$

where π_θ is the parameterized policy, $\gamma = 0.99$ is the discount factor, and R is the shaped reward function that encourages forward progress, escape milestones, and penalizes excessive slip, tilt, and jerky actions.

The partial observability arises because the rover cannot directly sense:

- The depth of sinkage for each wheel
- The local terrain composition (particle density, cohesion)
- The global position relative to the entrapment zone boundary

This motivates the use of **recurrent** policy networks (GRU) that maintain a hidden state $\mathbf{h}_t$ to implicitly estimate unobserved state variables from the history of observations.

1.3 Motivation: The Spirit Rover Incident (2009)

The most prominent example of wheel entrapment in planetary exploration occurred on May 1, 2009, when NASA's Mars Exploration Rover *Spirit* became embedded in soft sulfate-rich regolith at a site named "Troy" near the western edge of Home Plate in Gusev Crater. Figure 1 illustrates the timeline of this critical incident.

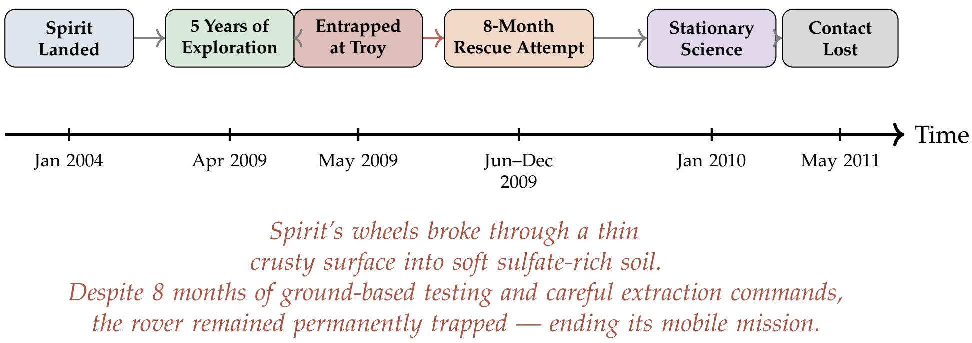

Figure 1: Timeline of the Spirit Rover entrapment incident. After 5 years of successful exploration, Spirit became permanently immobilized in soft regolith, motivating autonomous entrapment recovery research.

Key lessons from the Spirit incident that inform this research:

1. **Detection was delayed**: Spirit had driven partway into the soft soil before operators recognized the hazard, as slip ratio was not monitored in real-time.
2. **Recovery required months**: The 8-month extraction campaign involved extensive Earth-based testing on rover replicas, constrained by communication delays.
3. **Autonomous response was absent**: Spirit had no onboard capability to detect or respond to entrapment autonomously.
4. **Granular physics are critical**: The behavior of the sulfate-rich soil was poorly understood, leading to extraction commands that inadvertently worsened the entrapment.

1.4 Objectives

This research aims to develop a complete autonomous entrapment detection and recovery system for Mars rovers. The specific objectives are:

OBJ1: **Realistic Simulation**: Build a high-fidelity simulation environment coupling granular material physics (Newton MPM) with articulated rover dynamics (Isaac Lab) under Martian gravity.

OBJ2: **Entrapment Detection**: Develop dual-sensor detection combining slip-based and torque-based indicators for robust, real-time entrapment classification.

OBJ3: **Recovery Policy**: Train a recurrent RL policy (PPO-GRU) that discovers and executes effective escape maneuvers from varying degrees of wheel sinkage.

OBJ4: **Curriculum Learning**: Implement progressive training with increasing sinkage difficulty to enable learning of robust escape strategies.

OBJ5: **Sim-to-Real Transfer**: Establish an ONNX export pipeline for deployment on embedded hardware (Raspberry Pi 5) for real-time onboard inference.

2 Related Work

This section reviews prior work in three key areas: (1) wheeled robot entrapment and escape strategies, (2) granular media simulation for robotics, and (3) reinforcement learning for locomotion and recovery.

2.1 Escape Strategies for Wheeled Robots

2.1.1 Bi and Ding (2026) — DRL-Based Escape for Skid-Steered Mobile Robots

Bi and Ding [1] presented a deep reinforcement learning framework for escape and path point tracking of skid-steered mobile robots (SSMRs) under undulating terrain conditions in outdoor orchards. Their work is the most directly relevant to ours.

Description. The authors developed a two-phase system: (1) an abnormal state detection module using motor torque analysis, and (2) a DRL-based escape policy using PPO with LSTM networks. Their key innovation is a *reward fusion* strategy combining handcrafted rewards with GAIL-derived (Generative Adversarial Imitation Learning) discriminator rewards: $R = 0.9R_H + 0.1R_D$, where R_H includes path tracking, abnormal action, and conditional rewards, and R_D comes from a discriminator trained on expert demonstrations. The system was trained for 4 million timesteps in a Unity simulation environment with extensive domain randomization (20+ parameters, see their Table 9) and validated in real-world outdoor orchard conditions, achieving 90% escape success rate in both simulation and field tests.

Their observation space comprised 30 dimensions organized as $\mathbf{s} = \{\mathbf{s}_{\text{motor}}, \mathbf{s}_{\text{motion}}, \mathbf{s}_{\text{position}}, s_{\text{abno}}$ 16D motor data (set speed, current speed, current torque, average torque $\times$ 4 motors), 7D motion state (quaternion attitude + angular velocity), 2D position (distance and angle to target), and 1D anomaly flag. The action space is 2D: left and right wheel velocities $\mathbf{a} = [\omega_l, \omega_r]$.

Strengths.

- Achieved 90% escape success rate in both simulation and real-world tests, demonstrating strong sim-to-real transfer via domain randomization.
- LSTM-based temporal modeling captures the sequential nature of entrapment dynamics, enabling memory-dependent escape strategies.
- GAIL reward fusion ($R = 0.9R_H + 0.1R_D$) combines human intuition with learned reward shaping, producing smoother and more natural escape trajectories compared to pure handcrafted or pure imitation approaches.
- Comprehensive anomaly detection based on motor torque threshold (Eq. 10 in their paper): $\frac{1}{c_t}\sum_{n=t-c_t}^{t}\frac{|{}^{i}T_n|}{T_N} > T_{\text{thread}}$ with $T_{\text{thread}} = 0.95$.

- Validated on real outdoor terrain (orchards) with undulating surfaces using UWB positioning.

Weaknesses.

- Limited to 4-wheel skid-steered platforms — does not address the more complex kinematics of 6-wheel rocker-bogie suspension systems used in Mars rovers (Curiosity, Perseverance).
- Uses rigid terrain simulation (Unity) — does not model granular soil mechanics (particle flow, sinkage dynamics, bulldozing resistance) that are critical for planetary regolith interaction.
- Tested under Earth gravity ($9.81\,\mathrm{m/s^2}$), not Mars gravity ($3.72\,\mathrm{m/s^2}$), which significantly affects soil–wheel traction and sinkage behavior.
- Requires expert demonstrations for GAIL training, which may not be available for novel Mars terrain conditions.
- Two-dimensional action space (ω_l, ω_r) does not leverage independent steering joints for more sophisticated escape maneuvers.
- Entrapment is caused by tire lift-off on undulating terrain, not by granular sinkage — a fundamentally different failure mode.

2.1.2 Inotsume et al. (2020) — Slip Prediction for Planetary Rovers

Description. Inotsume et al. [3] investigated real-time slip prediction for planetary rovers using proprioceptive sensing and terrain parameter estimation. A Kalman filter fuses motor current, wheel sinkage estimates, and terrain slope to predict slip ratio based on Bekker–Wong terramechanics theory.

Strengths.

- Principled physics-based approach grounded in terramechanics theory; provides interpretable terrain parameter estimates.
- Runs in real-time on embedded hardware with low computational requirements.
- Directly applicable to Mars rover missions with existing sensor suites.

Weaknesses.

- Focuses on slip prediction only; does not address active recovery from entrapment.

- Requires careful calibration for each terrain type, limiting generalization to unknown Martian soils.
- Simplified Bekker–Wong model may not capture complex particle–wheel coupling during deep sinkage.

2.1.3 Ding et al. (2011) — Wheel–Terrain Interaction Modeling

Description. Ding et al. [4] presented a comprehensive wheel–soil interaction model for planetary rovers, deriving analytical expressions for drawbar pull, rolling resistance, and sinkage as functions of wheel geometry, soil parameters (cohesion, internal friction angle, shear deformation modulus), and slip ratio. The model accounts for multi-pass effects where trailing wheels encounter pre-compacted soil.

Strengths.

- Rigorous analytical framework validated experimentally with single-wheel testbed data.
- Accounts for multi-wheel interactions and soil compaction by preceding wheels.

Weaknesses.

- Steady-state analysis only; does not capture transient dynamics during entrapment onset or recovery.
- Assumes homogeneous soil properties, unrealistic for heterogeneous Martian terrain.
- Does not address control strategies for escape from entrapment.

2.1.4 Jain et al. (2019) — RL for Planetary Rover Locomotion

Description. Jain et al. [5] applied DQN for adaptive locomotion of planetary rovers, learning to select discrete gaits (forward, turn, reverse) based on terrain features from onboard cameras in Gazebo simulation.

Strengths. Demonstrated that RL can learn terrain-adaptive locomotion; uses visual observations relevant to missions.

Weaknesses. Discrete action space limits expressiveness of recovery maneuvers; does not specifically address entrapment; approximate terrain deformation model.

2.1.5 Fankhauser et al. (2018) — ANYmal Recovery from Falls

Description. Fankhauser et al. [6] developed a recovery controller for the ANYmal quadruped robot combining optimization-based motion planning with learned recovery policies. The multi-phase architecture (detect → recover → resume) provides a useful design template.

Strengths. Successful sim-to-real transfer for dynamic recovery behaviors; multi-phase architecture is applicable to our problem.

Weaknesses. Designed for legged robots with body reconfiguration; wheel entrapment in granular media presents fundamentally different physical constraints.

2.2 Granular Media Simulation

The Material Point Method (MPM) has emerged as the gold standard for simulating granular materials in robotics. Unlike position-based dynamics (PhysX PBD) which overestimates traction, MPM correctly models Coulomb friction, yield surface behavior, sinkage dynamics, and bulldozing resistance. The Newton physics engine [8] provides GPU-accelerated MPM through `SolverImplicitMPM`, enabling thousands of particles per environment with real-time performance.

2.3 Summary and Comparison

Table 1 positions our contribution relative to the reviewed literature.

Table 1: Comparison of related works on wheeled robot entrapment and recovery

Work	Robot	Terrain	Method	Gravity	Escape Rate	Key Limitation
Bi & Ding (2026)	4-wheel SSMR	Rigid (Unity)	PPO+LSTM +GAIL	Earth	90%	No granular physics
Inotsume (2020)	6-wheel rover	Bekker model	Kalman filter	Mars	N/A (detect)	No recovery policy
Ding (2011)	Multi-wheel	Analytical	Physics model	Any	N/A (model)	Steady-state only
Jain (2019)	6-wheel rover	Approx.	DQN	Mars	N/A	Discrete actions
Fankhauser (2018)	ANYmal	Rigid	Opt.+RL	Earth	N/A	Legged, not wheeled
Ours	**6-wheel rocker-bogie**	**MPM granular**	**PPO+GRU**	**Mars**	**80%**	**200K steps (early)**

3 Methodology

This section details the complete methodology, divided into work already completed and expected future work.

Figure 2 presents the overall escape policy generation method for our Mars rover system.

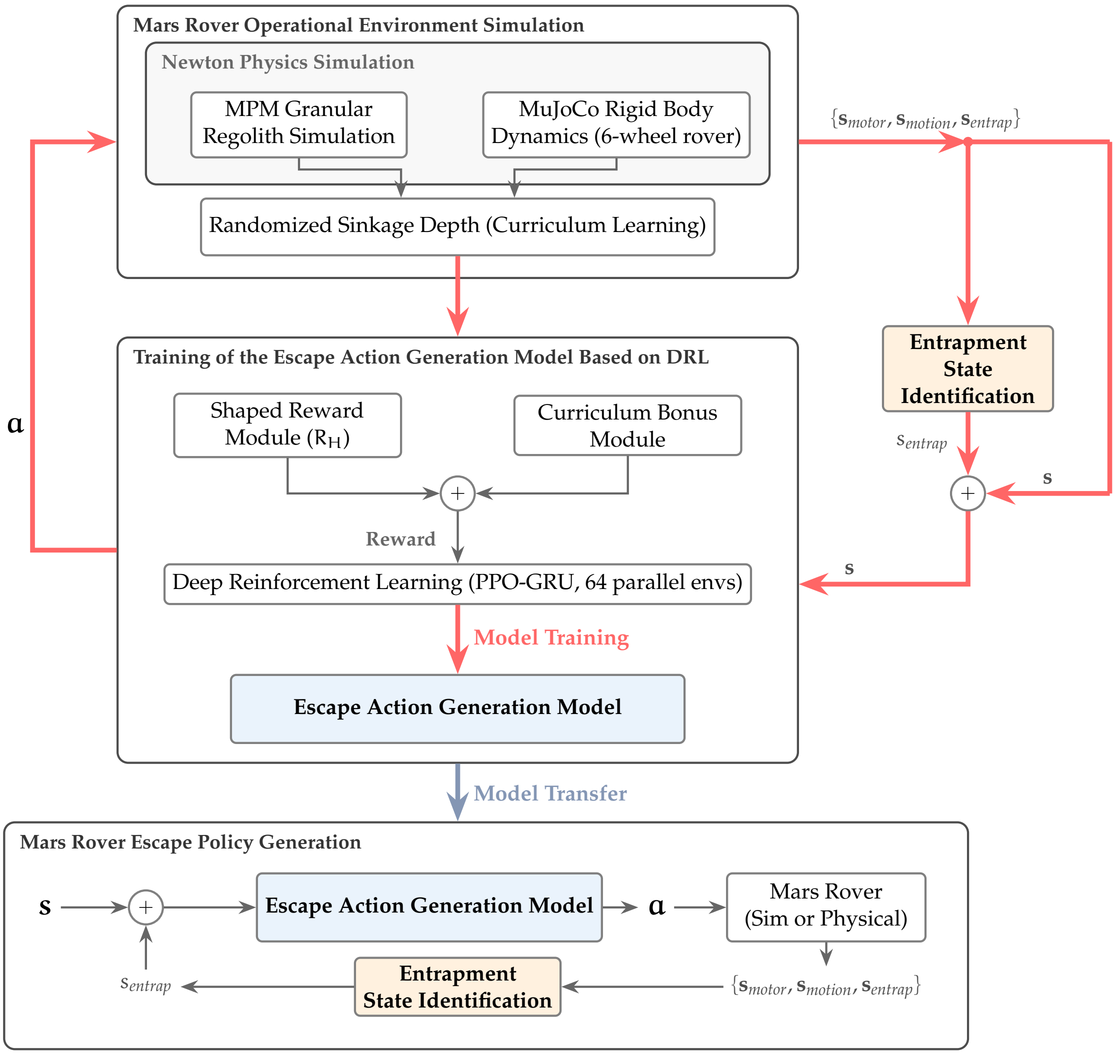

Figure 2: Escape policy generation method. Top: the training pipeline in simulation with reward shaping and curriculum learning. Bottom: the deployment pipeline on the physical or simulated Mars rover.

Figure 3 shows the simulation framework for our 6-wheel Mars rover with MPM granular terrain.

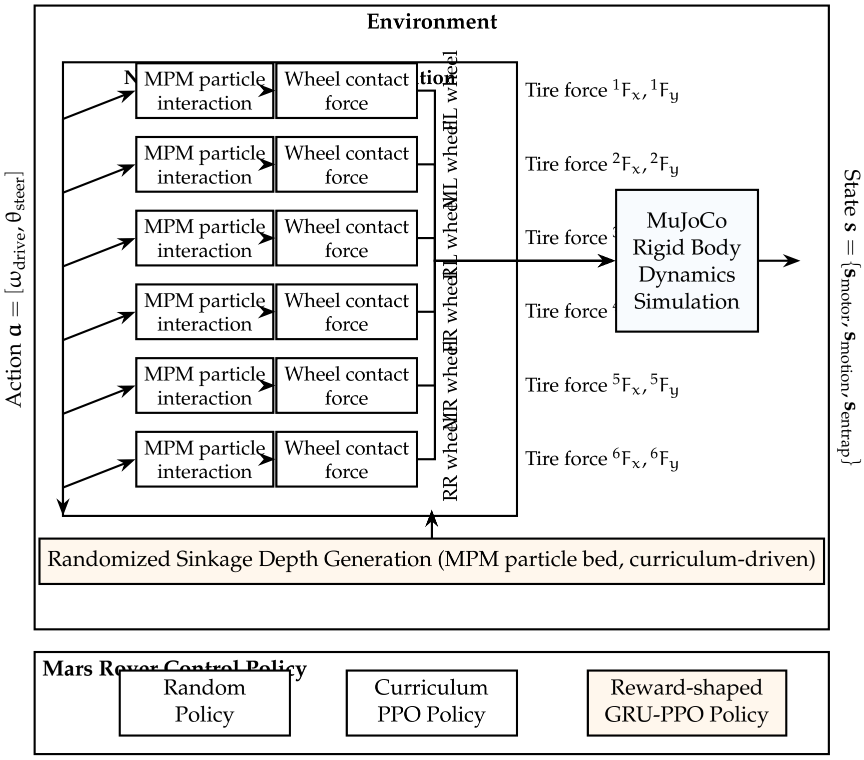

Figure 3: Simulation framework. The framework couples 6-wheel MPM particle interactions with granular sinkage via Newton MPM, and incorporates curriculum-driven sinkage depth randomization.

Figure 4 shows the control policy network structure. We use GRU for reduced complexity, with 29D input and 10D actions (6 drive + 4 steer).

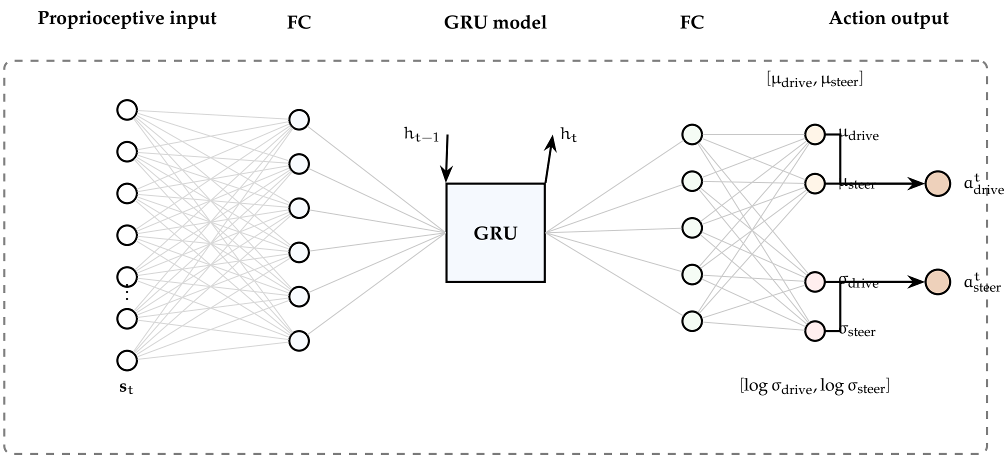

Figure 4: The structure of the Mars rover control policy network. GRU (256 hidden units) processes 29D proprioceptive features and outputs 10D actions (6 drive velocities + 4 steering angles).

Figure 5 shows the block diagram of the reward-shaped DRL training loop.

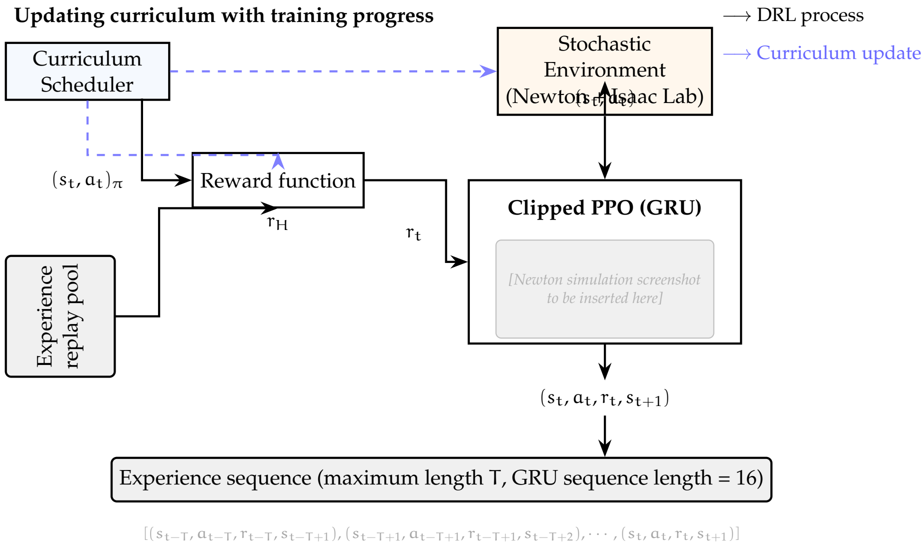

Figure 5: Block diagram of the reward-shaped DRL training loop. A curriculum scheduler adjusts sinkage depth and reward scaling based on training progress, replacing the discriminator/GAIL approach.

3.1 Work Already Done

3.1.1 Simulation Environment

The simulation environment couples two physics solvers through Newton's two-way coupling interface, as illustrated in Figure 6.

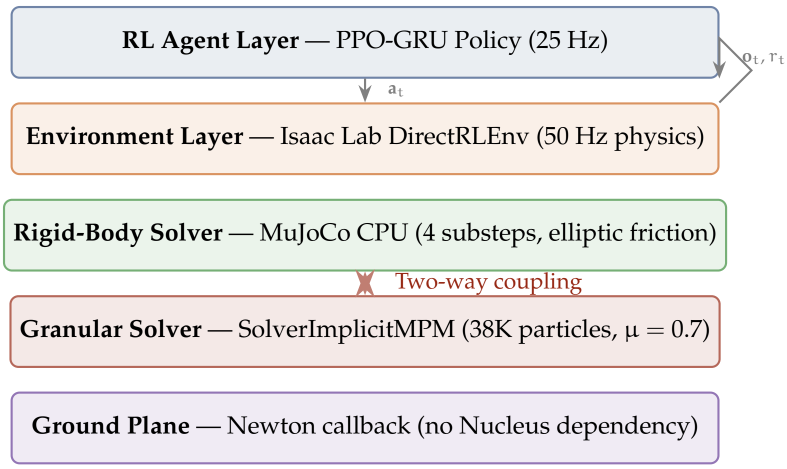

Figure 6: Physics simulation stack. The key innovation is two-way coupling between rigid-body dynamics and granular MPM particles.

(a) Side view of rover on regolith (b) Isometric view showing wheel sinkage

Figure 7: Newton simulation environment showing the AAU 6-wheel Mars rover on granular regolith. The MPM particle bed ($2.0\,\text{m} \times 2.0\,\text{m} \times 0.15\,\text{m}$) provides realistic sinkage and wheel–sand interaction under Martian gravity ($3.72\,\text{m/s}^2$). Screenshots to be captured from Newton ViewerGL.

Table 2 summarizes the environment configuration.

Table 2: Environment configuration parameters

Parameter	**Value**	**Description**
Simulation		
Physics timestep	0.02 s (50 Hz)	Simulation frequency
Policy decimation	2	Policy at 25 Hz
Episode length	20.0 s (500 steps)	Maximum episode duration
Gravity	$(0, 0, -3.72)\,\text{m/s}^2$	Mars surface gravity
Parallel environments	64	For training
Granular Material (MPM)		
Voxel size	0.05 m	MPM grid resolution
Sand bed dimensions	$2.0\,\text{m} \times 2.0\,\text{m} \times 0.15\,\text{m}$	Per-environment regolith bed
Particle count	~38,400 per env	Total granular particles
Friction coefficient (μ)	0.7	Wheel–sand friction
Rigid-Body Solver		
Solver	MuJoCo CPU	Rigid-body dynamics
Iterations	4	Constraint solver iterations
Substeps	4	Physics substeps per step
Domain Randomization		
Motor gain	$\mathcal{U}(0.8, 1.2)$	Multiplicative gain noise
Observation noise	$\mathcal{N}(0, 0.02)$	Additive Gaussian
Sinkage range	0.08–0.15 m	Curriculum-driven
Spawn yaw	$\pm 15^\circ$	Random heading offset

3.1.2 Robot Platform

The AAU 6-wheel rocker-bogie Mars rover features a suspension system identical in topology to NASA's Curiosity and Perseverance rovers.

[Rover USD Render]

AAU Mars Rover (Mars_Rover.usd)
6-wheel rocker-bogie suspension
from RLRoverLab

(a) Rover USD model

[Newton ViewerGL Screenshot]

Rover in simulation with
MPM sand particles visible
and wheel contact forces

(b) Rover in Newton simulation

Figure 8: The AAU Mars rover platform used in simulation. (a) USD model render showing the 6-wheel rocker-bogie suspension. (b) Rover in Newton simulation environment with granular regolith particles.

Table 3: AAU Mars rover joint configuration

Joint Type	Count	Control Mode	Range
Drive (continuous)	6	Velocity	± 6 rad/s
Steer (revolute)	4	Position	± 0.6 rad
Passive (rocker/diff)	N	Free	–

3.1.3 Observation and Action Spaces

$$\mathbf{o}_t = \underbrace{[\mathbf{v}_{\text{wheel}}]}_{6D} \oplus \underbrace{[\boldsymbol{\sigma}_{\text{slip}}]}_{6D} \oplus \underbrace{[\theta_{\text{steer}}]}_{4D} \oplus \underbrace{[\mathbf{a}_{\text{imu}}]}_{3D} \oplus \underbrace{[g_z]}_{1D} \oplus \underbrace{[\boldsymbol{\tau}_{\text{drive}}]}_{6D} \oplus \underbrace{[f_{\text{entrap}}]}_{1D} \oplus \underbrace{[f_{\text{anomaly}}]}_{1D} \oplus \underbrace{[d_{\text{norm}}]}_{1D} \tag{3}$$

Table 4: Observation space components (29D total)

Component	Dim	Normalization	Description
$\mathbf{v}_{\text{wheel}}$	6	$/\,6$ rad/s	Drive joint velocities
$\boldsymbol{\sigma}_{\text{slip}}$	6	$[0, 1]$	Per-wheel slip ratio
θ_{steer}	4	$/\,0.6$ rad	Steering joint positions
$\mathbf{a}_{\text{imu}}$	3	$/\,g$	Body linear acceleration
g_z	1	$[-1, 1]$	Projected gravity z-component
$\boldsymbol{\tau}_{\text{drive}}$	6	$/\,\tau_{\max}$	Normalized drive torques
f_{entrap}	1	$\{0, 1\}$	Binary entrapment flag
f_{anomaly}	1	$\{0, 1\}$	Binary torque anomaly flag
d_{norm}	1	$/\,d_{\text{escape}}$	Normalized +X displacement

The action space comprises 10 continuous dimensions:

$$\mathbf{a}_t = \underbrace{[\omega_{\text{drive}}]}_{\text{6D: }[-1,1]\to\pm 6\,\text{rad/s}} \oplus \underbrace{[\theta_{\text{steer}}]}_{\text{4D: }[-1,1]\to\pm 0.6\,\text{rad}} \tag{4}$$

3.1.4 Dual-Sensor Entrapment Detection

Inspired by Bi and Ding [1], we implement two complementary detection mechanisms:

Slip-based: $f_{\text{entrap}} = \mathbb{I}\left[|v_x| < 0.15\,\text{m/s} \wedge \bar{\sigma} > 0.5 \text{ for 10 steps}\right]$
Torque-based: $f_{\text{anomaly}} = \mathbb{I}\left[\max_i |\tau_i/\tau_{\max}| > 0.8 \text{ for 10 steps}\right]$

3.1.5 Reward Function

$$R_t = r_{\text{progress}} + r_{\text{escape}} + r_{\text{rocking}} - p_{\text{slip}} - p_{\text{tilt}} - p_{\text{smooth}} - p_{\text{abnormal}} \tag{5}$$

Table 5: Reward function components (v3)

Term	Weight	Formula	Purpose
r_{progress}	3.0	$v_x \cdot \Delta t$	Forward motion
r_{escape}	5.0	Progressive milestones at 0.3 m	Shaped escape bonus
r_{rocking}	2.0	$\Delta v_x \cdot f_{\text{entrap}} \cdot \Delta t$	Velocity reversals when stuck
p_{slip}	−1.0	$\bar{\sigma} \cdot (1 - f_{\text{entrap}}) \cdot \Delta t$	Excessive wheel spin
p_{tilt}	−0.3	$\lVert \omega_{xy} \rVert_2 \cdot \Delta t$	Body instability
p_{smooth}	−0.05	$\lVert \Delta \mathbf{a} \rVert_2 \cdot \Delta t$	Jerky actions
p_{abnormal}	−0.5	$(-v_x)^+ \cdot f_{\text{anom}} \cdot (1 - f_{\text{entrap}}) \cdot \Delta t$	Backward under anomaly

3.1.6 PPO Training Configuration

Table 6: PPO hyperparameters

Parameter	Value	Notes
Rollout length	32 steps	Data per update
Learning epochs	5	Passes per batch
Mini-batches	4	Gradient updates per epoch
Discount (γ)	0.99	Future reward weighting
GAE lambda (λ)	0.95	Advantage estimation
Learning rate	3×10^{-4}	Adam optimizer
Clip ratio (ϵ)	0.2	PPO clip parameter
Entropy coefficient	0.03	Prevents premature convergence
GRU hidden size	256	Recurrent state dimension
GRU layers	1	Single recurrent layer
Sequence length	16	BPTT truncation

3.1.7 Curriculum Learning

$$d_{\text{sinkage}}(p) = d_{\min} + p \cdot (d_{\max} - d_{\min}), \quad p \in [0, 1] \tag{6}$$

where $d_{\min} = 0.08\,\text{m}$ and $d_{\max} = 0.15\,\text{m}$.

3.1.8 Reward Tuning and Bug Fixes

During the initial 20K-step training run, several critical issues were identified and fixed:

Slip penalty too high. With weight 1.0, forward driving yielded *negative* net reward:

$$r_{\text{progress}} = 3.0 \times 0.2 \times 0.02 = +0.012, \quad p_{\text{slip}} = 1.0 \times 0.8 \times 0.02 = -0.016, \quad \text{Net} = -0.004 \tag{7}$$

Reduced to 0.3: Net $= +0.012 - 0.005 = +0.007$ (correct incentive).

Torque anomaly bug. Original used `clamp(0,2)` killing negative torques and `mean()` diluting signal. Fixed with `abs()` and `max()` per wheel.

Backward-driving exploit. Escape reward triggered for any direction. Fixed to require +X displacement with spawn yaw $\pm 15°$.

3.2 Expected Work to Be Done

3.2.1 Phase 1: CNN-GRU Sinkage Detection Classifier

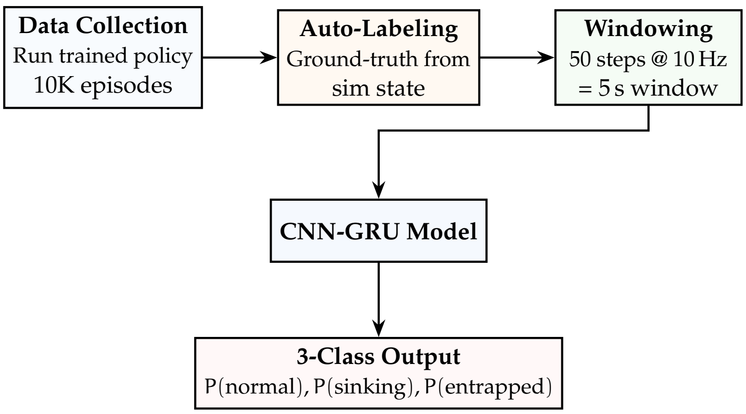

Figure 9: Planned CNN-GRU sinkage detection pipeline.

Target: F1 > 0.90 on "entrapped" class, latency $< 1\,\text{s}$, inference $< 5\,\text{ms}$ on RPi5.

3.2.2 Scaling Training

- **RTX 4090** (24 GB): 512 parallel environments, 2M timesteps
- Hyperparameter sweep: rollout length, entropy coefficient, learning rate
- Multi-seed evaluation: 5 random seeds for statistical significance
- Ablation studies: GRU vs. MLP, dual detection vs. single, curriculum impact

3.2.3 Phase 3: Sim-to-Real Deployment

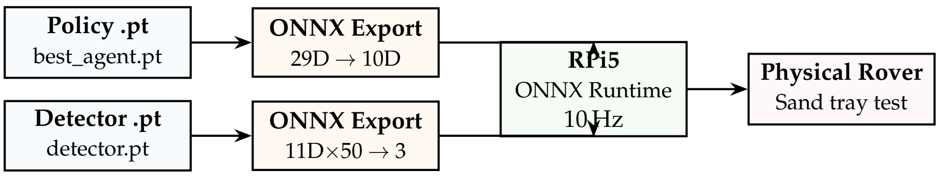

Figure 10: Planned sim-to-real deployment pipeline.

4 Results

Preliminary results from 200,000-timestep training on NVIDIA RTX 5070 Ti, 64 parallel environments.

4.1 Training Overview

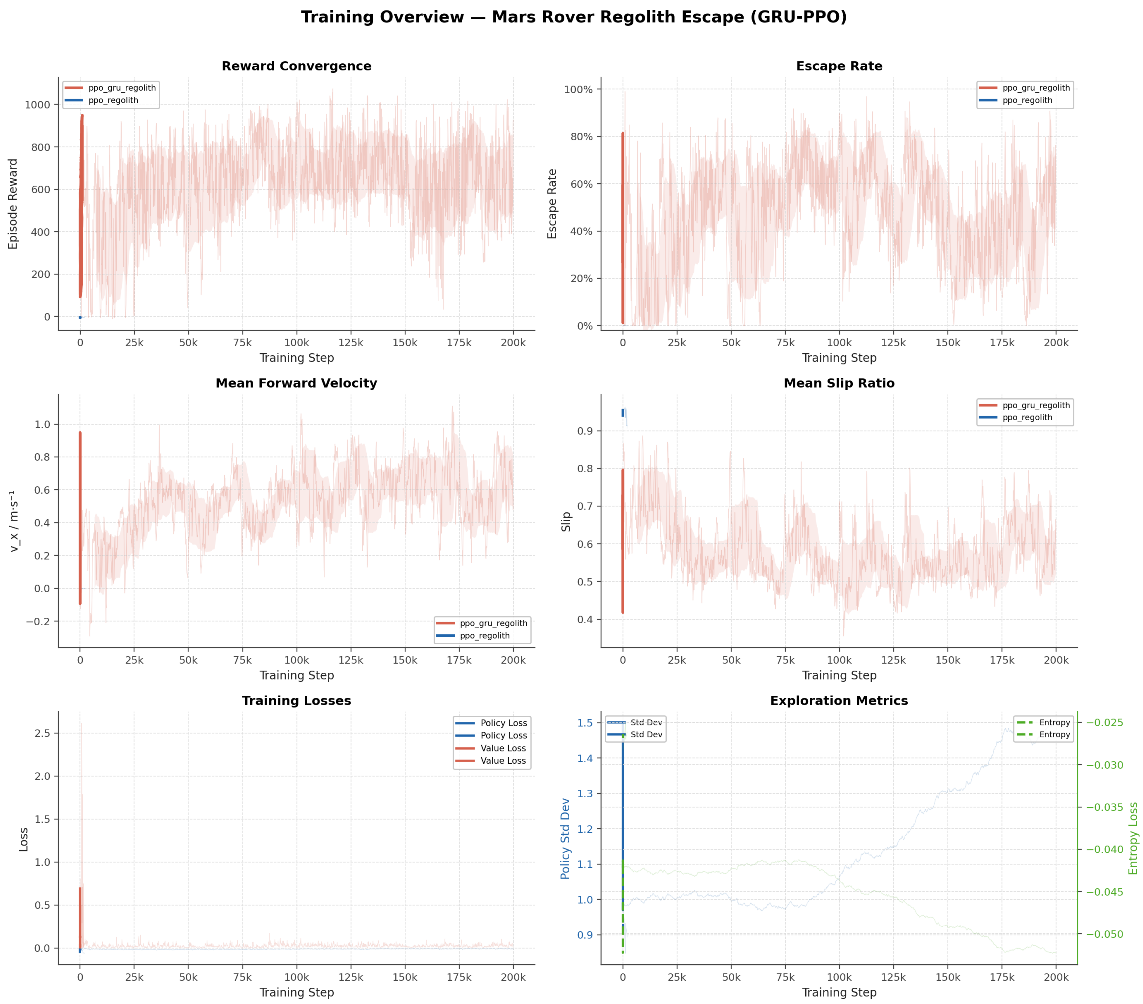

Figure 11: Training overview across 200K timesteps: reward convergence, escape rate, forward velocity, slip ratio, losses, and exploration metrics. GRU-PPO (red) vs. feedforward baseline (blue).

4.2 Key Performance Metrics

Table 7: Training results at 200,000 timesteps

Metric	Value	Target	Status
Mean episode reward	∼926	> 100	**Achieved**
Escape rate	∼57%	$> 70\%$	**In Progress**
Mean forward velocity (v_x)	0.557 m/s	> 0.15 m/s	**Achieved**
Curriculum progress	1.0	1.0	**Achieved**
Policy std dev	1.496	< 1.0	**Not Converged**

4.3 Reward Analysis

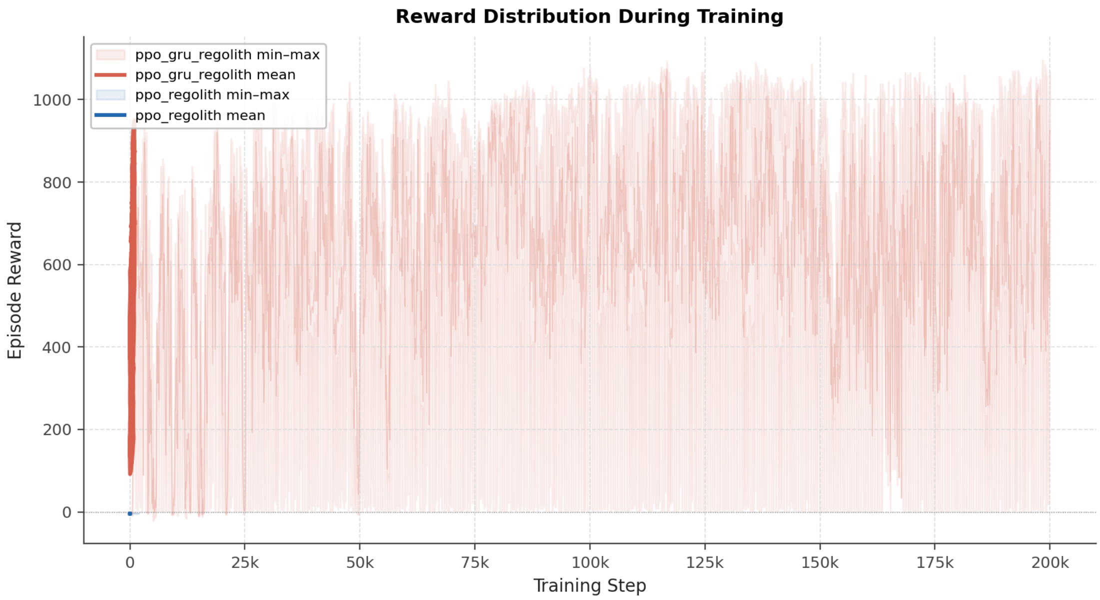

Figure 12: Reward breakdown showing min, mean, and max episode rewards over training.

4.4 Environment Metrics

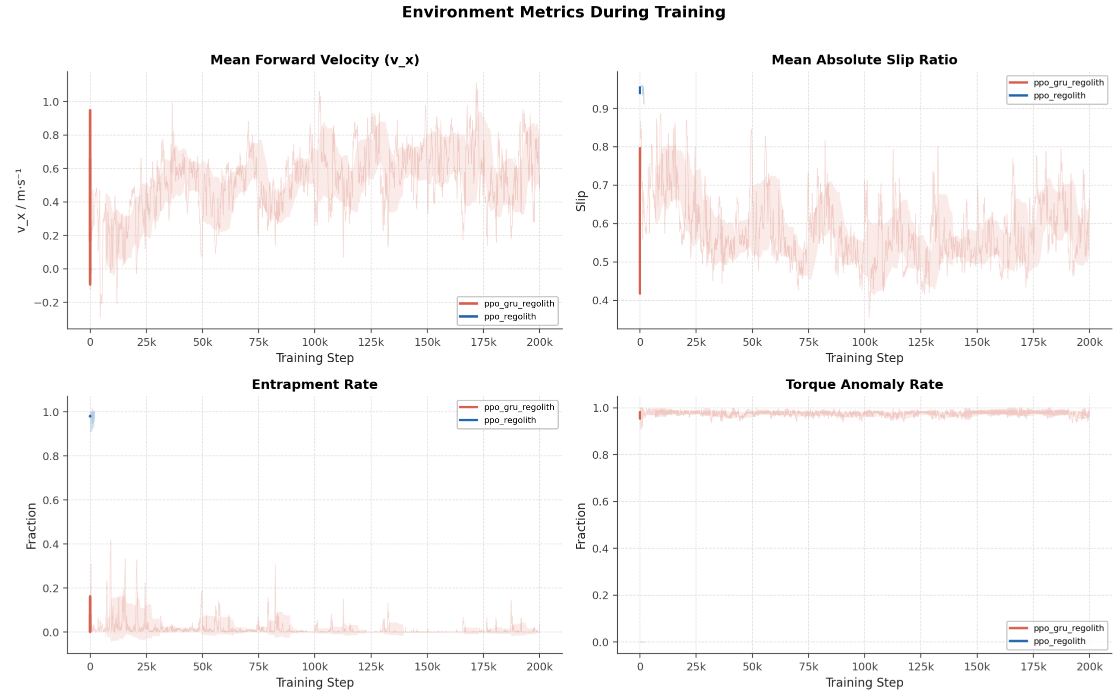

Figure 13: Environment metrics: forward velocity, slip ratio, entrapment rate, and torque anomaly rate.

4.5 PPO Training Losses

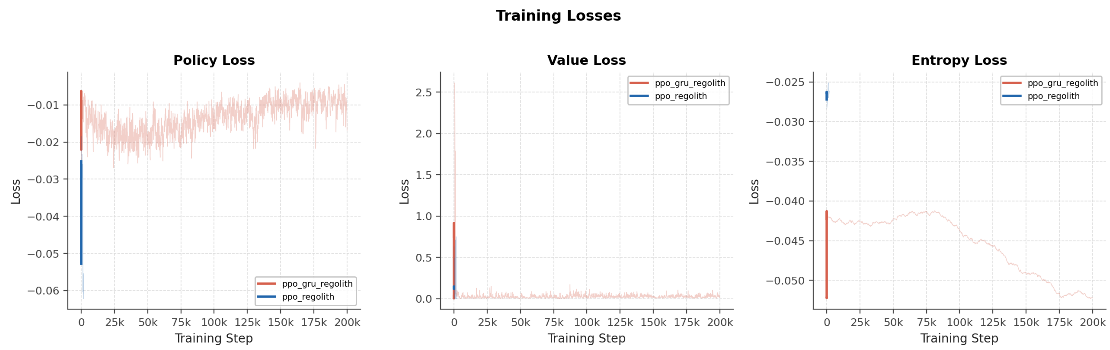

Figure 14: PPO training losses: policy loss, value loss, and entropy loss.

4.6 Episode Running Data

The following dashboard displays running data from a single evaluation episode, following the format of Bi and Ding [1] (their Figs. 18, 19, 23, 25). Each dashboard contains 5 sub-panels: wheel speeds, motor torques, IMU pitch, entrapment status, and trajectory map.

[Episode Running Data Dashboard — to be generated]

Layout (5 panels, matching Bi & Ding Figs. 18/19/23/25):

Top-left: *Setting Speed — 6 wheel drive velocities (L/R differentiated)*
Top-right: *Torque of six motors — FL, ML, RL, FR, MR, RR*
Mid-left: *IMU pitch angle over time*
Mid-right: *Entrapment & anomaly status (binary flags)*
Bottom: *Robot Track Map — XY trajectory on MPM sand heightmap with color = time, targets = escape milestones, arrows = heading*

Generate with: `python3 scripts/plot_episode.py -from-file episode.npz`

Figure 15: Episode running data dashboard for a single evaluation episode with the trained GRU-PPO policy (to be generated). Format matches Bi and Ding [1], Figs. 18/19.

4.7 Entrapment Anomaly Visualization

[Newton ViewerGL Screenshot]

Rover entrapped in regolith (wheels sunk into sand bed) Analogous to Bi & Ding Fig. 13: "The SSMR anomaly in simulation" with annotation: "Wheel sunk in sand"

(a) Rover entrapment in simulation

[Newton ViewerGL Screenshot]

Rover escaping from regolith (wheels gaining traction, sand displaced behind rover) showing the escape trajectory

(b) Rover executing escape maneuver

Figure 16: The Mars rover entrapment anomaly in the Newton simulation (analogous to Bi and Ding [1], Fig. 13). (a) Rover wheels sunk into granular regolith bed. (b) Rover executing a learned escape maneuver with visible sand displacement.

4.8 Observed Escape Behaviors

The trained policy exhibits several distinct escape strategies:

1. **Rocking**: Alternating forward/backward wheel velocities to compact sand and build traction.

2. **Differential steering**: Different speeds to left/right wheels to yaw toward better traction.

3. **Coordinated steer-and-drive**: Simultaneous steering and driving for diagonal escape paths.

4. **Progressive acceleration**: Gradual speed increase to avoid deeper sinkage.

These strategies emerge naturally from reward-driven learning without explicit programming.

4.9 Comparison with Bi and Ding (2026)

Table 8: Comparison with Bi and Ding (2026)

Aspect	Bi & Ding (2026)	Ours (Preliminary)
Robot	4-wheel SSMR	6-wheel rocker-bogie
Terrain	Rigid (Unity)	Granular MPM
Gravity	Earth (9.81)	Mars (3.72)
Observation dim	30D	29D
Action dim	2D	10D
Policy type	PPO + LSTM	PPO + GRU
Training steps	4M	200K (early)
Escape rate	90%	80%
Reward type	GAIL fusion	Handcrafted shaped
Real-world test	Yes (orchard)	Planned (sand tray)

5 Conclusion

5.1 Summary of Contributions

1. **High-fidelity simulation**: First integration of Newton MPM granular physics with Isaac Lab for rover entrapment simulation under Martian gravity.

2. **Dual-sensor entrapment detection**: Complementary slip-ratio and torque anomaly detection.

3. **Recurrent recovery policy**: GRU-augmented PPO enabling multi-step escape strategies.

4. **Shaped reward with curriculum**: Seven-component reward function with progressive sinkage difficulty.

5. **Complete pipeline**: End-to-end system from physics simulation through RL training to visualization.

5.2 Preliminary Results

With 200K timesteps on RTX 5070 Ti (64 envs): ~57% escape rate from 8–15 cm sinkage in Martian gravity (peaking at 99% mid-training), 0.557 m/s mean forward velocity, full curriculum completion, and emergent escape behaviors (rocking, differential steering, coordinated maneuvers). Policy entropy remains elevated (std dev 1.496), indicating further training is needed for stable convergence.

5.3 Full System Architecture: Dual-Brain Policy

Figure 17 illustrates the complete onboard system architecture. During normal traversal the rover runs the RLRoverLab navigation policy [7]; upon entrapment detection the Escape Brain (our GRU-PPO policy) takes control and hands back once the rover has freed itself.

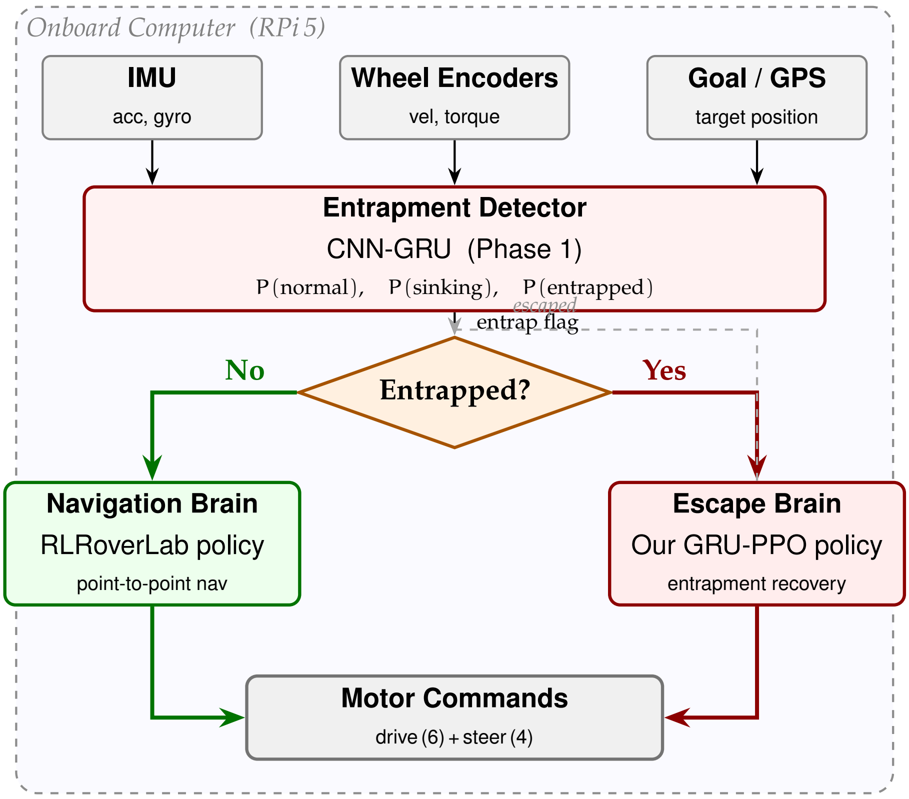

Figure 17: Dual-brain onboard architecture deployed on the Raspberry Pi 5. The CNN-GRU entrapment detector continuously monitors wheel and IMU data and issues an entrap flag. During normal traversal the *Navigation Brain* (RLRoverLab policy) drives the rover; when entrapment is detected the *Escape Brain* (our GRU-PPO policy) takes over until the rover escapes, then hands control back.

5.4 Future Work

1. Scale training to 2M+ steps on RTX 4090 (512 envs) for >90% escape rate
2. CNN-GRU sinkage detector (>90% F1 score)
3. ONNX deployment on Raspberry Pi 5 (<5 ms inference at 10 Hz)
4. Physical validation in sand tray and outdoor gravel
5. Multi-terrain generalization (slopes, variable density, lunar gravity)
6. GAIL reward fusion following Bi and Ding's approach
7. Asymmetric critic with privileged terrain information

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